S6 Proof Repair and Disproof Workbench — Private

Working mathematical notebook following the three-stream proof audit

26 August 2026

Abstract

This is a separate mathematical workbench for validating, repairing, or disproving the reconstructed compact complex threefold on S^6. The finite integer calculations are recorded as exact certificates: the monodromy invariant and coinvariant lattices, the finite-fibre indices, the parameter $p = 12\ell_0 - 4\ell_1 - 3\ell_2$ and Smith form $\mathrm{diag}(1, |p|)$, and the degree-two matrix with primitive left kernel $\Phi = (4, 2, 3, 2)$. The global conclusions remain **[CONDITIONAL]** until the printed matrices are identified with the actual geometric specialization, attaching, Mayer–Vietoris, and Leray maps.

The notebook names those bridge lemmas (C1–C3, V1, MV1, and L1), gives a conditional recognition theorem, and treats the Section 10/CDP issue in three separate questions: is a printed proof step valid, is the published theorem false, and does this candidate satisfy its hypotheses? It does not silently alter the reconstructed manuscript; the frozen 121-page audit and 108-page transcription remain under `audit_baseline/`.

For the Section 10 route, compact-Cartier-fibre duality and the required top-degree analytic base-change theorem are now proved here under explicit hypotheses. Properness and flatness are automatic once the upstream construction really supplies a compact connected smooth threefold and a nonconstant map to the curve; identifying the displayed singular surface with the full scheme-theoretic fibre remains candidate-specific.

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1 Problem, evidence boundary, and status language

The target is the manuscript's claimed compact complex threefold X equipped with a holomorphic map $f\colon X \to \mathbb{P}^1$, with

$$X \cong_{\mathrm{diff}} S^6, \qquad a(X) = 1, \qquad \langle c_3(T_X), [X] \rangle = 2.$$

The review found no finite-algebra contradiction in the completed manuscript, but it did not certify the geometric inputs from which the terminal matrices are obtained.

1.1 Status rules

[VERIFIED] Reproduced exactly or proved in this notebook from stated hypotheses.

[CONDITIONAL] Correct after explicitly named geometric or analytic hypotheses are granted.

[OPEN] A load-bearing statement for which no complete proof or counterexample is yet recorded here.

[REFUTED] A statement defeated by a printed counterexample or a complete contradiction.

1.2 Current ledger

Status	Object	Present evidence
[VERIFIED]	terminal finite algebra	Monodromy products/orders, Smith forms, exterior-power ranks, displayed lattice saturation, period identities, $\eta^2 = 12$, and the Gram determinant have exact checks in the retained review harnesses.
[VERIFIED]	compact-Cartier duality	Ramis–Ruget duality, its Gorenstein complete-intersection calculation, Cartier adjunction, and compact coherent-cohomology finiteness give the stated perfect pairing without a normality hypothesis.
[VERIFIED]	top-degree base change	The analytic cohomology-and-base-change theorem gives β_c^2 as an isomorphism when H^3 of the fibre vanishes; compactness, smoothness, connectedness, and nonconstancy make properness and flatness automatic over a smooth curve.
[CONDITIONAL]	$\pi_1(X)$ and $H_*(X;\mathbb{Z})$	Correct for the printed presentations and boundary matrices; their equality with the maps induced by the analytic gluings is not yet established here.

Status	Object	Present evidence
[OPEN]	cusp construction	Full chartwise proof of the $\bar{\Lambda}$-action, proper discontinuity, quotient, central fibre, and global equivariant collapse.
[OPEN]	integral topology maps	Specialization, transfer indices, saturation, torsion, primitive bases, orientations, and degree-three signs.
[CONDITIONAL]	candidate duality application	The abstract duality theorem is proved, but applying it here still requires the reconstructed W to be the compact full scheme-theoretic Cartier fibre in the asserted smooth threefold.
[OPEN]	Section 10 completion	The abstract duality and base-change implications are proved. The candidate construction and fibre identification, conductor/Picard hypotheses, and a line-by-line comparison with CDP remain to be established.

The working rule is asymmetric: a failed proof step does not refute the theorem, and a coherent terminal calculation does not prove that the required geometric map exists.

2 Fixed algebraic data reproduced by the audits

This section freezes the exact statements that need no further numerical experimentation. The research task is to prove that the geometric maps have these algebraic representatives.

Let

$$H_1(F;\mathbb{Z}) = \Lambda = \langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle, \qquad H^1(F;\mathbb{Z}) = V = \langle \gamma, u, w, \delta \rangle.$$

Proposition 2.1 (checked monodromy lattices)**.** ***[VERIFIED]*** *The displayed monodromy matrices satisfy*

$$A_1 A_2 M_0 = I, \qquad \ker(M_0 - I) = \operatorname{im}(M_0 - I) = \Lambda_{\mathrm{tor}} := \langle \widehat{w}, \widehat{\delta} \rangle.$$

For $T_0 = I + N$ with $N^2 = 0$, the invariant lattices in degrees one through three are

$$\begin{aligned} V^{T_0} &= \langle \gamma, u \rangle, \\ (\wedge^2 V)^{T_0} &= \langle \gamma u, \gamma\delta, uw, \gamma w - u\delta \rangle, \\ (\wedge^3 V)^{T_0} &= \langle \gamma uw, \gamma u\delta \rangle. \end{aligned}$$

They are saturated and have ranks $(1,2,4,2,1)$ across all degrees. For $G = \langle T_1, T_2 \rangle$, the nonzero coinvariant lattices used downstream are

$$V_G = \mathbb{Z}\bar{\delta}, \qquad (\wedge^2 V)_G = \mathbb{Z}[\gamma\delta], \qquad (\wedge^3 V)_G = \mathbb{Z}[uw\delta],$$

all torsion-free, with $[uw] = 6[\gamma\delta]$.

Proposition 2.2 (checked finite-fibre indices)**.** ***[VERIFIED]*** *In degrees $q = 0,1,2,3,4$, the two finite-fibre specialization-index tuples in the manuscript's bases are*

$$(1,3,1,1,3), \qquad (1,4,2,2,4).$$

For $v_2 = -\varepsilon'$, the degree-three index-two quotient at p_2 is generated by $[\gamma uw]$.

Proposition 2.3 (checked group presentation)**.** ***[VERIFIED]*** *Once the asserted based inclusions are granted, the presentation reduces to*

$$\left\langle c, x, y \;\middle|\; c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2} \right\rangle.$$

After eliminating y, its relation matrix is

$$R = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det R = -p, \qquad p = 12\ell_0 - 4\ell_1 - 3\ell_2,$$

and its Smith form is $\operatorname{diag}(1, |p|)$.

Proposition 2.4 (checked degree-two terminal matrix)**.** ***[VERIFIED]*** *The displayed degree-two matrix has columns*

$$(2,-4,-2,3),\ (1,-2,-2,3),\ (3,0,-4,0),\ (0,1,0,-1),\ 0,\ 0.$$

Its image has rank three and is saturated, its cokernel is $\mathbb{Z}$, and its primitive left-kernel generator is

$$\Phi = (4, 2, 3, 2).$$

Proposition 2.5 (checked all-degree terminal homology package)**.** ***[VERIFIED]*** *For $p \neq 0$, the retained exact audit applied to the manuscript's printed degree-by-degree Mayer–Vietoris/Leray data gives the terminal abelian-group tuple*

$$(H_0, H_1, H_2, H_3, H_4, H_5, H_6) \cong (\mathbb{Z}, \mathbb{Z}/|p|\mathbb{Z}, \mathbb{Z}/|p|\mathbb{Z}, \mathbb{Z}/|p|\mathbb{Z}, \mathbb{Z}/|p|\mathbb{Z}, 0, \mathbb{Z}).$$

Consequently the printed algebraic package has the homology of S^6 when $p = -1$. The exact evidence is frozen in `audit_baseline/annotated_audit.tex`, ledger item B-EX-06, and in the degree-by-degree calculation surrounding Theorem 7.22 of `audit_baseline/clean_transcription.tex`.

Every proposition in this section concerns the printed matrices and lattices. None identifies those matrices with a geometric specialization, attaching, Mayer–Vietoris, or Leray map.

3 The prior Fable certificate: relevance and boundary

The separate Erdős–Strauss workspace contains a pinned copy of Jon Poplett's JGPTech package for the polynomial map publicly attributed there to Fable. This is relevant as a standard of verification, not as evidence for any step of the present S^6 construction. The canonical checkout and its immutable mirror are listed with exact absolute paths in `comparative_context/README.md`; within either checkout the relevant file is `JGPTech_Fun/Jacobian/aceg.py`.

Define $F = (F_1, F_2, F_3)\colon \mathbb{C}^3 \to \mathbb{C}^3$ by

$$\begin{aligned} F_1(x,y,t) &= (1+xy)^3 t + y^2(1+xy)(4+3xy), \\ F_2(x,y,t) &= y + 3x(1+xy)^2 t + 3xy^2(4+3xy), \\ F_3(x,y,t) &= 2x - 3x^2 y - x^3 t. \end{aligned}$$

Proposition 3.1 (replayed finite certificate)**.** ***[VERIFIED]*** *Exact rational-polynomial arithmetic gives*

$$\det DF = -2.$$

Moreover, the three distinct points

$$\left(0, 0, -\tfrac{1}{4}\right), \qquad \left(1, -\tfrac{3}{2}, \tfrac{13}{2}\right), \qquad \left(-1, \tfrac{3}{2}, \tfrac{13}{2}\right)$$

all map to $(-\frac{1}{4}, 0, 0)$. The local pinned checker `Jacobian/aceg.py` was replayed successfully on 26 August 2026 with the 'base' and manifest checks. Its exact SHA-256, mirror commit, and absolute paths are recorded in `comparative_context/README.md`. Thus the displayed map itself has the advertised constant nonzero Jacobian and noninjective fibre.

Remark 3.2 (provenance boundary). The located corpus is an independently organized proof and exact audit of the displayed map, not an original model-session transcript. No original Fable/Claude model-session source was located. None of this certificate supplies a lemma about the complex-threefold manuscript, and the Jacobian counterexample is kept distinct from Erdős–Strauss arithmetic.

The transferable lesson is procedural. The prior result terminates in a small certificate—one determinant identity and one explicit collision—that can be checked independently of its discovery narrative. The present paper will reach the same evidentiary level only when its analytic gluings terminate in independently reconstructed integral chain maps. Accordingly this workbench uses typed claims, logical dependency edges, typed morphisms, explicit uncertainties, clean-archive replays, and rendered-page QA; see `comparative_context/README.md`.

4 Topology: exact reduction and geometric obligations

4.1 What the exact arithmetic already proves

Let

$$p = 12\ell_0 - 4\ell_1 - 3\ell_2.$$

The retained exact harness verifies the determinant and Smith-normal-form calculations for the printed relation matrix. In particular, the manuscript's chosen parameters $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ give $p = -1$.

Proposition 4.1 (terminal group calculation)**.** ***[VERIFIED]*** *For the printed based presentation and its stated meridian relations, the resulting group is cyclic of order $|p|$. Hence the printed presentation is trivial for $(0, 1, -1)$.*

Proof. The exact row and column operations recorded in the audit reduce the displayed 2×2 relation matrix to Smith normal form $\mathrm{diag}(1, |p|)$. A group presented by that matrix is therefore $\mathbb{Z}/|p|\mathbb{Z}$, which is trivial when $p = -1$. □

Remark 4.2. Proposition 4.1 is a theorem about the printed presentation. It becomes a theorem about $\pi_1(X)$ only after the based inclusions induced by all three analytic attaching maps have been derived in the same generators and orientations.

4.2 The map that must be identified

Write the relevant decomposition schematically as $X = U \cup N$, where N is the union of the filling pieces and $U \cap N$ is the disjoint union of their collars. The homological boundary map is

$$\delta_q = (i_{U*}, -i_{N*}) \colon H_q(U \cap N; \mathbb{Z}) \longrightarrow H_q(U; \mathbb{Z}) \oplus H_q(N; \mathbb{Z}).$$

The printed matrices denoted α_q have verified ranks, Smith forms, and saturation properties. The central question is not another Smith-form computation; it is the equality

$$\boxed{\alpha_q = [\delta_q]_{\mathcal{B}_q}}$$

for explicitly constructed primitive oriented integral bases $\mathcal{B}_q$.

The equality is decomposed below into the six nonredundant proof obligations C1, C2, C3, V1, MV1, and L1. The first three construct the finite equivariant model, compute its integral collapse map, and identify that map with nearby-cycle specialization. Obligation V1 fixes the based van Kampen inclusions; MV1 identifies the actual Mayer–Vietoris boundary cylinders and multiple-fibre indices; L1 supplies an independently oriented Leray–Mayer–Vietoris comparison. This crosswalk is exhaustive for the conditional endpoint stated here; none of the six is replaced by matching a rank or by choosing the desired matrix as a normalization.

Disproof test 4.3 (fast falsification targets)**.** Any of the following disproves the claimed topology calculation: a nonprimitive geometric image where α_q assumes a primitive one; an index or torsion class omitted by specialization; a meridian sign changing $p = -1$ to a nonunit; or a Leray transgression differing from $\pm p\omega$ by a nonunit factor.

4.3 Conditional recognition

Theorem 4.4 (precise conditional endpoint)**.** *[CONDITIONAL] Assume obligations C1–C3, V1, MV1, and L1 are discharged and give exactly the printed integral maps. Then the verified terminal calculations yield*

$$\pi_1(X) = 0, \qquad H_k(X;\mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & 1 \le k \le 5. \end{cases}$$

Consequently X is a smooth homotopy 6-sphere; the standard fact $\Theta_6 = 0$ identifies it diffeomorphically with S^6.

Proof. The hypotheses identify the actual van Kampen and Mayer–Vietoris/Leray maps with the already verified presentations and matrices. Their Smith forms give the displayed fundamental group and homology. A simply connected integral homology sphere is a homotopy sphere by Hurewicz and Whitehead; the final smooth recognition uses $\Theta_6 = 0$. □

5 First proof attacks: exact lemmas to establish or break

The following statements turn the review's broad objections into falsifiable mathematical targets. A proof must derive the maps from the analytic charts; matching ranks is insufficient.

5.1 Cusp quotient and collapse

The source defines an infinite toric threefold Y from the height-one cones over the periodic A_2 triangulation and an action of the form

$$\Psi_{\bar\lambda}(x,t) = (c_{\bar\lambda}(t)t^{B_0\bar\lambda}x, t), \qquad \bar\lambda \in \bar\Lambda.$$

Proof obligation 5.1 (C1: finite equivariant cusp model)**.** Construct a locally finite $\bar\Lambda$-equivariant CW model of $(Y_{\le\eta}, Y_0)$ whose quotient is finite, and prove that the collapse is equivariant in the diagram

$$\begin{array}{ccc} Y_{\le\eta} & \xrightarrow{r_Y} & Y_0 \\ \big\downarrow{\scriptstyle /\bar\Lambda} & & \big\downarrow{\scriptstyle /\bar\Lambda} \\ N_0' & \xrightarrow{r} & W. \end{array}$$

The proof must include freeness and proper discontinuity of the full action, continuity across the one-, two-, and three-branch strata, compatibility with the fibre-torus action, and a strong deformation retraction.

Proof obligation 5.2 (C2: integral collapse matrix)**.** With fixed cellular orientations, prove for $0 \le q \le 4$ the exact sequence

$$0 \longrightarrow \operatorname{im}(\wedge^q M_0 - I) \longrightarrow \wedge^q\Lambda \xrightarrow{c_*} H_q(W;\mathbb{Z}) \longrightarrow 0.$$

In particular, derive $c_*\colon \Lambda \to H_1(W) = \bar\Lambda$ as the quotient map, prove

$$c_*(\widehat\gamma \wedge \widehat u) = [\mathbb{T}_B], \qquad c_*(\widehat w \wedge \widehat\delta) = 0,$$

and record the degree-three orientations rather than inferring them from ranks.

Proof obligation 5.3 (C3: collapse equals specialization)**.** Prove functorially and integrally that

$$c^* = i_F^* r^* = \mathrm{sp}_{\mathrm{nc}}^q, \qquad \operatorname{im} c^* = (\wedge^q V)^{T_0},$$

including saturation, torsion, bases, and the degree-three sign. The comparison diagram must state the precise nearby-cycle edge map; the equality may not be defined by choosing the manuscript's answer as a normalization.

5.2 Based inclusions

Choose one basepoint on the zero section and fixed whiskers through every collar.

Proof obligation 5.4 (V1: primitive meridians and drift)**.** Derive from the coordinate gluings, with no unrecorded conjugations,

$$\widehat{g}_j^{\,m_j} = t_{v_j}, \qquad \widetilde{a}_0 = t_\mu \widehat{\rho}_2^{-1} \widehat{\rho}_1^{-1}.$$

Prove that the clockwise meridian lifts to $\widehat{g}_j$ rather than its inverse or a nonprimitive power, that the two finite-fibre signs agree with the chosen orientations, that the zero-section whiskers add no hidden translation, and that the tautological cusp disc gives $\mu = 0$.

5.3 Mayer–Vietoris and Leray

Proof obligation 5.5 (MV1: geometric boundary cylinders)**.** Using the independently constructed collars and the maps from C2–C3, prove for every relevant degree q that

$$\alpha_q = \left[(i_{U*}, -i_{N*}) : H_q(U \cap N; \mathbb{Z}) \longrightarrow H_q(U; \mathbb{Z}) \oplus H_q(N; \mathbb{Z})\right]_{\mathcal{B}_q}$$

in the manuscript's named primitive oriented bases. This must include a chain-level check of every zero column, sign, torsion class, and saturation claim; equality of ranks or Smith forms is not sufficient. For $K_k = \ker(H_k(M) \to H_k(W))$, derive geometrically

$$K_1 = \mathbb{Z}[\widetilde{a}_0], \qquad K_2 = \langle \xi_{\widehat{w}}, \xi_{\widehat{\delta}} \rangle,$$

and fix one orientation convention for

$$\partial(j_* \xi_\nu) = (I - A_1^{-1})\nu = (I - A_2)\nu.$$

The same calculation must recover the multiple-fibre class $[F] \longmapsto (3[S_1], -4[S_2])$ and the index-two degree-three image at p_2.

Proof obligation 5.6 (L1: transgression comparison)**.** Construct the clutching cocycle that compares the Leray transgression with the Mayer–Vietoris obstruction. For 12γ, derive the discrepancy triple

$$(4\ell_1 - \alpha_1,\ 3\ell_2 - \alpha_2,\ -(12\ell_0 - \alpha_1 - \alpha_2)),$$

whose sum is $-p$, and thereby determine—rather than assume—the sign in

$$d_2^{0,1}(12\gamma) = \pm p\, \omega.$$

Then prove that the cup-product and coinvariant identifications are natural in the same orientations before using multiplicativity in higher degrees.

Until C1–C3, V1, MV1, and L1 are established, the conclusion $X \cong S^6$ remains conditional. The integer algebra downstream of these statements is already fixed by the previous section.

6 The non-normal fibre: exact claim under investigation

Status convention. This notebook does not assert that the CDP theorem is false. It studies one specific inference in the printed proof and the conditional application of a conductor-supported differential to the candidate fibre W. A defect in the printed inference, an alternative proof of the published theorem, and failure of the candidate construction are logically distinct possibilities.

6.1 The local conductor proposition

Proposition 6.1 (conductor-supported ambient differential)**.** ***[CONDITIONAL]*** *Let $g : \mathcal{X} \to \Delta$ be proper, with $\mathcal{X}$ a smooth complex threefold, and let $S = g^{-1}(0)$ be a reduced normal-crossings surface. Write $\eta : \widetilde{S} \to S$ for the normalization, $D \subset S$ for the reduced singular locus, and $\widetilde{D} = \eta^{-1}(D)$. Let A be a line bundle on S such that $\eta^* A \simeq \mathcal{O}_{\widetilde{S}}$, and choose a nowhere-vanishing section e of $\eta^* A$. Then there is a nonzero section*

$$s \in H^0\Big(S, \Omega^1_{\mathcal{X}}|_S \otimes A\Big)$$

whose image in $\Omega^1_S \otimes A$ is supported on D, hence is torsion, and whose image in

$$\widetilde{\Omega}^1_S \otimes A, \qquad \widetilde{\Omega}^1_S := \Omega^1_S / \mathcal{T}(\Omega^1_S),$$

is zero. If S is connected and $A \not\simeq \mathcal{O}_S$, then $H^0(S, N^*_{S/\mathcal{X}} \otimes A) = H^0(S, A) = 0$, so s does not come from the conormal term.

Proof. Locally $g = ux$, uxy, or $uxyz$, where u is a unit. The conormal map is generated by $dg|_S$. At a double point,

$$dg|_S = u(y\,dx + x\,dy),$$

and at a triple point it has coefficients in (xy, xz, yz). Thus $dg|_S$ is nonzero away from D but restricts to zero on D; equivalently, it lies in the conductor ideal times $\Omega^1_{\mathcal{X}}|_S$.

The normalization–conductor square gives

$$0 \longrightarrow \mathcal{O}_S \longrightarrow \eta_* \mathcal{O}_{\widetilde{S}} \oplus \mathcal{O}_D \longrightarrow \eta_* \mathcal{O}_{\widetilde{D}} \longrightarrow 0.$$

After tensoring with the locally free sheaf $E = \Omega^1_{\mathcal{X}}|_S \otimes A$, the compatible pair

$$(\eta^*(dg|_S) \otimes e,\ 0)$$

descends uniquely to s. It is nonzero off D. Its intrinsic differential vanishes off D, so its image in $\Omega^1_S \otimes A$ is conductor-supported torsion. For nontrivial A, a global section of A pulls back to constants on the compact connected components of $\widetilde{S}$. Compatibility across the conductor and connectedness of S force either all constants to vanish or all to be nonzero; the latter would trivialize A. Hence $H^0(S, A) = 0$. □

Consequently the implication

$$\Big[H^0(S, N^*_{S/\mathcal{X}} \otimes A) = 0, \quad H^0(S, \widetilde{\Omega}^1_S \otimes A) = 0\Big] \Longrightarrow H^0(S, \Omega^1_{\mathcal{X}}|_S \otimes A) = 0$$

is not a formal consequence of the conormal sequence for a non-normal normal-crossings surface. The official CDP article prints precisely this reduction on p. 692 [2, p. 692]. This observation concerns that inference, not by itself the truth value of the theorem.

6.2 Conditional application to the candidate fibre

[CONDITIONAL] Assume the following upstream assertions about the candidate:

(i) $f : X \to B$ is a proper nonconstant holomorphic map from a smooth threefold to a smooth curve;

(ii) $W = f^{-1}(p_0)$ is the reduced scheme-theoretic fibre and is a normal-crossings Cartier divisor;

(iii) $\eta : \widetilde{W} \to W$ has $\widetilde{W} \simeq \mathrm{dP}_6$;

(iv) $H^2(X;\mathbb{Z}) = 0$.

For any $L \in \mathrm{Pic}(X)$, put

$$i : W \hookrightarrow X, \qquad \mathcal{F}_L = T_X \otimes L, \qquad A_L = (L^\vee \otimes \omega_X)|_W.$$

Because $H^2(X;\mathbb{Z}) = 0$, $c_1(L^\vee \otimes \omega_X) = 0$. Since $H^1(\mathrm{dP}_6, \mathcal{O}) = 0$, the Chern-class map $\mathrm{Pic}(\mathrm{dP}_6) \to H^2(\mathrm{dP}_6;\mathbb{Z})$ is injective. Therefore

$$\eta^* A_L \simeq \mathcal{O}_{\widetilde{W}}.$$

The conductor proposition yields

$$H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0.$$

6.3 The exact duality implication required

Proposition 6.2 (absolute duality on a compact Cartier fibre)**.** ***[VERIFIED]*** *Let $f : X \to B$ be a holomorphic map from a smooth complex threefold to a smooth complex curve. Suppose that $i : W = f^{-1}(b) \hookrightarrow X$ is compact, is the full scheme-theoretic fibre, and is an effective Cartier divisor. For every line bundle L on X, put*

$$\mathcal{F}_L = T_X \otimes L, \qquad A_L = (L^\vee \otimes \omega_X)|_W.$$

Then W is a pure two-dimensional Gorenstein complex space and there is a functorial perfect pairing

$$H^2(W, i^*\mathcal{F}_L) \times H^0(W, \Omega^1_X|_W \otimes A_L) \longrightarrow \mathbb{C}.$$

Equivalently,

$$H^2(W, i^*\mathcal{F}_L)^\vee \simeq H^0(W, \Omega^1_X|_W \otimes A_L).$$

No normality hypothesis on W is required.

Proof. Because W is an effective Cartier divisor in the smooth threefold X, each local ring of W is a two-dimensional hypersurface quotient of a regular local ring. Thus W is Cohen–Macaulay and Gorenstein. If $K^\bullet_W$ denotes the analytic dualizing complex, the Gorenstein criterion and complete-intersection computation of Ramis–Ruget give

$$K^\bullet_W \simeq \omega_W[2]$$

for an invertible dualizing sheaf ω_W [4, Proposition 2 and the complete-intersection calculation, pp. 90–91].

Cartier adjunction follows by applying $\mathcal{H}om_X(-, \omega_X)$ to

$$0 \longrightarrow \mathcal{O}_X(-W) \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_W \longrightarrow 0 :$$

$$\omega_W = \mathcal{E}xt^1_X(i_*\mathcal{O}_W, \omega_X) \simeq (\omega_X \otimes \mathcal{O}_X(W))|_W.$$

Since $W = f^*(b)$ as a scheme-theoretic Cartier divisor, $\mathcal{O}_X(W) \simeq f^*\mathcal{O}_B(b)$. Its restriction to W is the pullback of the one-dimensional vector space $\mathcal{O}_B(b)|_b$, so a choice of nonzero vector gives

$$\mathcal{O}_X(W)|_W \simeq \mathcal{O}_W, \qquad \omega_W \simeq \omega_X|_W.$$

The choice changes the final pairing by a nonzero scalar only.

Ramis–Ruget duality pairs, for coherent $\mathcal{G}$,

$$H^p_c(W,\mathcal{G}) \quad \text{with} \quad \operatorname{Ext}^{-p}_W(\mathcal{G}, K^\bullet_W)$$

[4, Theorem 2, pp. 82–83]. Compactness gives $H^p_c(W,\mathcal{G}) = H^p(W,\mathcal{G})$. With $p=2$ and $\mathcal{G} = i^*\mathcal{F}_L$, the shift above gives

$$\operatorname{Ext}^{-2}_W(\mathcal{G}, K^\bullet_W) \simeq \operatorname{Hom}_W(\mathcal{G}, \omega_W).$$

The coherent cohomology groups are finite-dimensional by Grauert finiteness [7]; in particular their natural topologies are separated, so the strong duality of Ramis–Ruget is an ordinary algebraic perfect duality. Since $\mathcal{G}$ is locally free,

$$\begin{aligned}\operatorname{Hom}_W(\mathcal{G},\omega_W) &= H^0(W, \mathcal{G}^\vee \otimes \omega_W)\\ &\simeq H^0(W, \Omega^1_X|_W \otimes L^\vee|_W \otimes \omega_X|_W)\\ &= H^0(W, \Omega^1_X|_W \otimes A_L).\end{aligned}$$

The pairing is cup product followed by evaluation and the trace $H^2(W,\omega_W) \to \mathbb{C}$. Precomposition on the Hom term shows the required contravariance in $\mathcal{G}$. □

Corollary 6.3 (the duality bridge used in Section 10)**.** *Under the hypotheses of the proposition,*

$$H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0 \quad \Longrightarrow \quad H^2(W, i^*\mathcal{F}_L) \neq 0.$$

[CONDITIONAL] The corollary applies to the reconstructed candidate only after its upstream construction proves that X is smooth near W and that W really is the compact full scheme-theoretic Cartier fibre. Under those hypotheses, the conductor section gives

$$H^2(W, i^*\mathcal{F}_L) \neq 0.$$

[VERIFIED] The abstract duality theorem is proved.
[CONDITIONAL] Its candidate-specific application still depends on the upstream construction. The next subsection separates the equally standard abstract base-change theorem from its unproved candidate-specific hypotheses.

6.4 The exact top-degree base-change implication required

Proposition 6.4 (top-degree analytic base change)**.** ***[VERIFIED]*** *Let $f : X \to B$ be a proper morphism of complex spaces, $\mathcal{F}$ a coherent sheaf flat over B near $c \in B$, and X_c the analytic, hence full scheme-theoretic, fibre. If $H^3(X_c, \mathcal{F}|_{X_c}) = 0$, then the canonical map*

$$\beta^2_c : (R^2 f_*\mathcal{F})_c \otimes_{\mathcal{O}_{B,c}} \mathbb{C}(c) \xrightarrow{\sim} H^2(X_c, \mathcal{F}|_{X_c})$$

is an isomorphism. In particular, a nonzero target forces $(R^2 f_*\mathcal{F})_c \neq 0$.

Proof. Bănică–Stănăşilă's base-change theorem identifies surjectivity of the degree-q fibre map with right exactness of the corresponding direct-image functor. Their top-degree corollary states that if $H^q(X_c, \mathcal{F}|_{X_c}) = 0$, then these equivalent conditions hold in degree $q-1$, and the degree-$(q-1)$ base-change maps are isomorphisms [6, Chapter III, §3, Theorem 3.4 and Corollary 3.5, pp. 116–121]. Taking $q = 3$ gives the displayed map. If its target is nonzero, its source is nonzero; hence the stalk itself cannot vanish. □

Lemma 6.5 (automatic properness and flatness over the curve)**.** ***[VERIFIED]*** *Let X be a compact connected smooth complex manifold, let B be a smooth complex curve, and let $f : X \to B$ be nonconstant and holomorphic. Then f is proper and flat. Every locally free $\mathcal{O}_X$-module is f-flat, and every full scheme-theoretic fibre is an effective Cartier divisor of pure codimension one.*

Proof. The underlying continuous map is proper because X is compact and B is Hausdorff. Fix $x \in X$, put $b = f(x)$, and write $A = \mathcal{O}_{B,b}$ and $R = \mathcal{O}_{X,x}$. The ring A is a discrete valuation ring and R is a regular local analytic ring, hence an integral domain. The homomorphism $A \to R$ is injective: if a nonzero germ on the curve pulled back to zero near x, then a neighbourhood of x would map into its discrete zero set, so f would be locally constant and therefore, by connectedness and analytic continuation, globally constant.

Thus R is torsion-free as an A-module. A torsion-free module over a discrete valuation ring is flat, so f is flat at x; since x was arbitrary, f is flat. A locally free sheaf has stalk R^r, which is flat over A by the same argument. Finally, if t is a local parameter at b, its pullback is a nonzero element of the domain R and hence a non-zero-divisor. Therefore the full fibre is locally cut out by one non-zero-divisor: it is an effective Cartier divisor and is pure of codimension one. □

For a full fibre of complex dimension two and a coherent sheaf, $H^3 = 0$. Thus, once the reconstructed object is independently shown to be a compact connected smooth threefold, f is shown to be nonconstant, and the displayed W is identified with its full scheme-theoretic fibre, the lemma makes f proper and flat and makes the locally free sheaf $\mathcal{F}_L = T_X \otimes L$ f-flat. The proposition then gives

$$(R^2 f_* \mathcal{F}_L)_{p_0} \otimes \mathbb{C}(p_0) \xrightarrow{\sim} H^2(W, i^* \mathcal{F}_L).$$

Combined with the conditional conductor and duality calculation, this would imply

$$(R^2 f_* \mathcal{F}_L)_{p_0} \neq 0, \qquad \text{hence} \qquad R^2 f_*(T_X \otimes L) \neq 0.$$

Proof obligation 6.6 (candidate base-change instantiation)**.** **[OPEN]** Prove from the reconstructed charts that X is a compact connected smooth complex threefold, that f is a nonconstant holomorphic map, and that the displayed W is exactly its full scheme-theoretic fibre at p_0. The lemma then supplies properness, flatness, Cartierness, and f-flatness of $\mathcal{F}_L$; the proposition supplies the base-change isomorphism. This obligation is only the candidate-specific construction check. Do not replace it with semicontinuity alone.

6.5 Present evidentiary status

- The local conductor calculation was independently checked and called sound by Reviewers A and C.
- Reviewer A found the application plausible but requested exact duality and base-change hypotheses and citations. The present notebook now supplies the two abstract theorems and isolates their candidate-specific construction hypotheses; this post-audit work has not yet received an independent external adjudication.
- Reviewer B classified the corresponding lemma and theorem as secondary and reviewed them only for consistency; Reviewer B did not independently certify the analytic bridge.
- Reviewer C called the application coherent and the printed-proof issue substantiated, while explicitly separating it from falsity of the published theorem and from validity of the candidate construction.
- The application to W remains conditional on the upstream construction of X, the asserted fibre geometry, and $H^2(X; \mathbb{Z}) = 0$, none of which the three audits collectively certify.

6.6 Verification and falsification programme

1. **Local algebra.** Work in $\mathbb{C}\{x,y,z\}/(xy)$ and $\mathbb{C}\{x,y,z\}/(xyz)$. Verify the conductor ideals, injectivity of $\mathcal{O}_S\,dg \to \Omega^1_{\mathcal{X}}|_S$, exactness of the normalization–conductor sequence at double and triple points, and the non-conormal character of s.

2. **Independent reconstruction of W.** Verify directly from the quotient charts that the full fibre is reduced, Cartier and normal-crossings, that its normalization is exactly dP_6, and that the stated conductor curves and triple points are exhaustive.

3. **Picard calculation.** Recompute $\eta^*((L^\vee \otimes \omega_X)|_W)$ from explicit transition functions, rather than only from the claimed global topology. A single L for which this pullback is nontrivial would falsify the universal application of the conductor proposition.

4. **Candidate duality hypotheses.** The abstract pairing is proved above. Verify from the candidate charts that W is compact and is the full Cartier fibre in the asserted smooth threefold, then independently identify $(T_X \otimes L)|^\vee_W \otimes \omega_W$. Failure of any candidate hypothesis prevents application of the verified theorem.

5. **Base-change audit.** Verify the upstream compactness, connectedness, smoothness, nonconstancy, and full-fibre assertions. The automatic-flatness lemma then discharges properness, flatness, Cartierness, and f-flatness before the stated degree-3-to-degree-2 theorem is applied. Alternatively, construct the local finite-free complex representing $Rf_*\mathcal{F}_L$ and compute $\beta^2_{p_0}$ directly.

6. **Literature comparison.** Reproduce the CDP p. 692 argument with exactly the same fibre, twist, and definition of $\widetilde{\Omega}^1_S$. Determine whether an omitted hypothesis restores the inference or whether an independent proof bypasses it. Either outcome must be recorded without converting a proof-gap diagnosis into a theorem verdict.

Proof obligation 6.7 (CDP comparison)**.** [OPEN] Produce a line-by-line diagram comparing the candidate's ambient, conormal, and intrinsic cotangent sequences with the exact sheaves in the cited CDP reduction. Record separately: the validity of the printed inference, the possibility of an independent repair, and the validity of the candidate construction.

Disproof test 6.8 (decisive outcomes)**.** The Section 10 route fails if the descended section s is zero after the true conductor identifications, if it lies in the conormal image, if the candidate fibre is not compact, full, and Cartier in the asserted smooth threefold, or if the candidate does not satisfy the hypotheses needed to instantiate the verified top-degree base-change theorem. Conversely, verifying the candidate-fibre hypotheses and proving proof obligations 6.6 and 6.7 would remove the printed CDP reduction as an automatic contradiction, but would still leave the construction and topology obligations of C1–C3, V1, MV1, and L1.

7 Confirmed local repairs kept separate from the theorem

These corrections are mathematically definite but do not settle the central construction.

1. [REFUTED] Proposition 4.7(iv): the printed closedness justification is not valid as written. A repair would require a complete chartwise/local-finiteness argument, which remains [OPEN].

2. [REFUTED] Remark 7.28 is false for a fixed triple. For example,

$$(\ell_0,\ell_1,\ell_2) = (100,1,-1), \qquad p = 12(100) - 4(1) - 3(-1) = 1199 \neq \pm 1.$$

   The intended statement must quantify existentially over ℓ_0 for fixed admissible (ℓ_1,ℓ_2).

3. [VERIFIED] Appendix Lemma A.3 prints cohomological arrows while labelling them as homology. The sequence must be written

$$H^k(W) \longrightarrow H^k(\widetilde{W}) \oplus H^k(D) \longrightarrow H^k(\widetilde{D}) \longrightarrow H^{k+1}(W),$$

followed by universal coefficients when homology ranks are required.

4. [VERIFIED] Appendix Remark A.6 must read $\pi_1(N_0) = \bar{\Lambda} = \pi_1(W)$, not $\pi_1(N_0) = \Lambda$.

5. [VERIFIED] The modular assembler has a page-71 portability defect under the tested MiKTEX route; the canonical monolithic source renders correctly. This is a build defect, not a mathematical counterexample.

8 Research log

Round 0 — private workbench initialization

- Preserved the completed 121-page annotated audit as `audit_baseline/annotated_audit.tex` and the 108-page reconstruction as `audit_baseline/clean_transcription.tex`.
- Converted the review objections into named mathematical obligations with explicit maps, bases, signs, and falsification tests.
- Proved the exact conditional endpoint: once the geometric maps are identified with the printed matrices, the verified algebra gives the standard smooth 6-sphere.
- Kept the conductor mechanism separate from both the existence of X and the truth or falsity of the published CDP theorem.

Next calculations

1. Extract the actual coordinate formulas for the three collar inclusions and build a complete orientation/sign table.
2. Build a finite integral chain model for the cusp central fibre W and compare its specialization map with Appendix B in primitive bases.
3. Compute the Leray d_2 geometrically and compare it to $d_2(12\gamma) = \pm p\omega$ without importing the Mayer–Vietoris normalization.
4. Write the exact local normalization–conductor diagram for Section 10 and audit every duality/base-change hypothesis.

Round 1 — 26 August 2026: typed workbench release

- Rebuilt the workbench as a new clean tree with the exact title "S6 Proof Repair and Disproof Workbench — Private". The public audit project is not an input to the upload workflow and remains untouched.
- Kept the terminal algebra [VERIFIED], but kept the global topology [CONDITIONAL] until C1–C3, V1, MV1, and L1 identify the printed matrices with actual geometric maps. No theorem-level CDP verdict is inferred from a proof-step defect alone.

- Added a reader-facing literature audit with DOI-resolvable references for CDP20/CDP98, analytic duality and base change, nearby cycles, topology, and the Leiden Declaration. Local SGA TeX was used as a comparison layer; third-party files are not bundled.
- Added the bounded Fable/Jacobian provenance question to the comparative workstream. Any source path not independently located remains **[OPEN]** rather than being presented as a recovered transcript.

Round 2 — 26 August 2026: duality and base-change bridges

- Proved the abstract compact-Cartier-fibre duality statement from Ramis–Ruget's analytic dualizing complex and trace pairing, Cartier adjunction, and coherent-cohomology finiteness. Normality of the fibre is not required.
- Promoted only the abstract theorem to **[VERIFIED]**. Its application to the reconstructed fibre remains **[CONDITIONAL]** on the candidate-specific smoothness, compactness, full scheme-theoretic fibre, and Cartier assertions.
- Proved the required abstract top-degree base-change isomorphism from the degree-3 vanishing corollary of the analytic base-change theorem. Also proved that properness, flatness, Cartierness of the full fibre, and f-flatness of a vector bundle are automatic for a nonconstant map from a compact connected smooth complex manifold to a smooth curve.
- Kept only the candidate instantiation **[OPEN]**: the reconstructed charts must actually produce that compact connected smooth threefold, that nonconstant map, and the displayed full scheme-theoretic fibre.
- Reconciled the conditional topology endpoint with the six exact bridge obligations C1–C3, V1, MV1, and L1; no four-item shorthand now substitutes for that dependency set.

9 Literature audit and source gate

This is a publication-facing literature record, not a list of private chat notes. The CDP comparison is against Campana–Demailly–Peternell's 2020 paper [2, Theorem 2.2 and Corollary 2.3, p. 680], with the 1998 paper [3, Lemma 3.2, pp. 84–85] retained because the later argument imports its semidirect-product calculation. The exact theorem and page locators are given in the bibliography below; the DOI links are the reader-facing source gate.

For the analytic part, the workbench uses the absolute duality results of Ramis–Ruget [4, Theorems 1–2 and Proposition 2, pp. 81–91] and the relative analytic reference of Ramis–Ruget–Verdier [5], together with Bănică–Stănășilă's cohomology-and-base- change criteria [6]. The latter is cited at Chapter III, §3, Theorems 3.4 and Corollaries 3.5 and 3.10, and §4, Theorem 4.12. The Stacks Project is used only as an algebraic cross-check, not as a substitute for the analytic theorem. Nearby-cycle formalism is compared with SGA 7 I/II and with the analytic treatment cited in the bibliography; none of those general references identifies the candidate's printed matrices without a candidate-specific map calculation.

The local literature corpus was also checked before writing this section. The relevant SGA 6 and SGA 7 TeX paths are recorded verbatim in `research/README.md` and `comparative_context/README.md`; they were read as a local comparison layer. No third-party source files or PDFs are copied into this project. The public citations remain the canonical way for a reader to check the literature.

The mathematical role of each source is deliberately narrow:

- CDP20 supplies the theorem-level target and the published proof steps under audit; CDP98 supplies the earlier split-extension calculation.

- Ramis–Ruget, Ramis–Ruget–Verdier, and Bănică–Stănășilă supply the analytic duality and base-change statements whose hypotheses are still checked explicitly in Section 10.
- SGA 7 and the analytic nearby-cycle references supply formal frameworks [21, 22], not the candidate-specific integral identifications.
- Hatcher, Smale, and Kervaire–Milnor supply the final recognition theorems [9, 11, 12] only after the geometric maps have been certified.
- The Leiden Declaration is recorded as a methodological disclosure reference [1]; it is not mathematical evidence for any claim.

No literature result is silently promoted from a general theorem to a fact about the candidate. Any such promotion must appear as a typed claim in the ledgers and carry its own map, hypothesis, and falsification test.

References

[1] J. Alper, M. Barany, A. Chavarri Villarello, S. Dahmen, W. Dean, K. Ganapathy, M. Harris, D. Holmes, M. Jamnik, S. Kelk, B. Kra, U. Martin, B. Naskręcki, R. Ochigame, J. Portegies, and J. Schmitt, *Leiden Declaration on Artificial Intelligence and Mathematics*, 2026, doi:10.5281/zenodo.20302944.

[2] F. Campana, J.-P. Demailly, and T. Peternell, *The algebraic dimension of compact complex threefolds with vanishing second Betti number*, Compos. Math. **156** (2020), 679–696, doi:10.1112/S0010437X19007802.

[3] F. Campana, J.-P. Demailly, and T. Peternell, *The algebraic dimension of compact complex threefolds with vanishing second Betti number*, Compos. Math. **112** (1998), 77–91, doi:10.1023/A:1000313214795; arXiv:math/9607215.

[4] J.-P. Ramis and G. Ruget, *Complexe dualisant et théorèmes de dualité en géométrie analytique complexe*, Publ. Math. IHÉS **38** (1970), 77–91, doi:10.1007/BF02684652.

[5] J.-P. Ramis, G. Ruget, and J.-L. Verdier, *Dualité relative en géométrie analytique complexe*, Invent. Math. **13** (1971), 261–283, doi:10.1007/BF01406078; EuDML record.

[6] C. Bănică and O. Stănăşilă, *Algebraic Methods in the Global Theory of Complex Spaces*, Editura Academiei and John Wiley & Sons, Bucharest and London, 1976; Chapter III, §3, Theorem 3.4 and Corollaries 3.5 and 3.10, pp. 116–123; Chapter III, §4, Theorem 4.12, pp. 134–135, publicly accessible scan.

[7] H. Grauert, *Ein Theorem der analytischen Garbentheorie und die Modulräume komplexer Strukturen*, Publ. Math. IHÉS **5** (1960), 5–64, doi:10.1007/BF02684746.

[8] J. M. Lee, *Introduction to Complex Manifolds*, Graduate Studies in Mathematics **244**, American Mathematical Society, Providence, RI, 2024; Theorem 9.37, publisher page.

[9] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002; Theorems 4.5 and 4.32 and Corollary 4.33, author's electronic edition.

[10] G. E. Bredon, *Sheaf Theory*, second edition, Graduate Texts in Mathematics **170**, Springer, New York, 1997; Chapter VI, doi:10.1007/978-1-4612-0647-7.

[11] S. Smale, *Generalized Poincaré's conjecture in dimensions greater than four*, Ann. of Math. (2) **74** (1961), 391–406, doi:10.2307/1970239.

[12] M. A. Kervaire and J. W. Milnor, *Groups of homotopy spheres: I*, Ann. of Math. (2) **77** (1963), 504–537, doi:10.2307/1970128.

[13] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics **52**, Springer, New York, 1977.

[14] C. Birkenhake and H. Lange, *Complex Tori*, Progress in Mathematics **177**, Birkhäuser, Boston, 1999; Chapter 2, §6, doi:10.1007/978-1-4612-1566-0.

[15] G. Pareschi, M. Popa, and C. Schnell, *Hodge modules on complex tori and generic vanishing for compact Kähler manifolds*, Geom. Topol. **21** (2017), 2419–2460; §8, p. 2437, doi:10.2140/gt.2017.21.2419.

[16] W. Fulton, *Introduction to Toric Varieties*, Annals of Mathematics Studies **131**, Princeton University Press, Princeton, 1993.

[17] D. A. Cox, J. B. Little, and H. K. Schenck, *Toric Varieties*, Graduate Studies in Mathematics **124**, American Mathematical Society, Providence, RI, 2011; Theorem 12.1.10, doi:10.1090/gsm/124.

[18] V. I. Danilov, *The geometry of toric varieties*, Russian Math. Surveys **33** (1978), no. 2, 97–154; §9, doi:10.1070/RM1978v033n02ABEH002305.

[19] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics **7**, Springer, New York, 1973; Chapter VII, doi:10.1007/978-1-4684-9884-4.

[20] K. S. Brown, *Cohomology of Groups*, Graduate Texts in Mathematics **87**, Springer, New York, 1982.

[21] A. Grothendieck, M. Raynaud, and D. S. Rim (eds.), *Groupes de monodromie en géométrie algébrique. I*, Séminaire de Géométrie Algébrique du Bois-Marie 1967–1969, Lecture Notes in Mathematics **288**, Springer, Berlin, 1972, doi:10.1007/BFb0068688.

[22] P. Deligne and N. Katz (eds.), *Groupes de monodromie en géométrie algébrique. II*, Séminaire de Géométrie Algébrique du Bois-Marie 1967–1969, Lecture Notes in Mathematics **340**, Springer, Berlin, 1973.

[23] A. Dimca, *Sheaves in Topology*, Universitext, Springer, Berlin, 2004; Definition 4.2.1, Propositions 4.2.2 and 4.2.11, Theorem 2.3.26, and Corollary 2.3.4, doi:10.1007/978-3-642-18868-8.

[24] M. Kashiwara and P. Schapira, *Sheaves on Manifolds*, Grundlehren der mathematischen Wissenschaften **292**, Springer, Berlin, 1990; §8.6, doi:10.1007/978-3-662-02661-8.

[25] A. Grothendieck and the SGA contributors, *SGA in English: maintained working edition*, 2026, doi:10.5281/zenodo.21925497; source repository. This working edition is cited only as a searchable reading layer; the original volumes remain the authority.

[26] The Stacks Project Authors, *The Stacks Project*, stacks.math.columbia.edu, especially Tags 0AUW and 0EX2.